

Adithya V

✉ adith348@gmail.com ☎ +91 98458 95271 in vadithya 🌐 itsadithyav

Summary

Computer Science Engineer specializing in **Generative AI**, **Computer Vision**, and **System Reliability**. Offers a dual background in building experimental AI models (Llama 3.1, TensorFlow) and executing rigorous QA for consumer electronics (LG WebOS). Proven ability to fine-tune LLMs to outperform benchmarks and ensure production-grade system stability.

Experience

Manual Test Engineer *LG Soft India* Apr 2025 – July 2025

- Managed Quality Assurance for **WebOS** on 4K/8K display modules, executing **400+** test cases across 10+ external devices to ensure 100% launch compatibility.
- Identified and reported **9 significant bugs**, including critical crashes in high-traffic apps like **Netflix** and **Chrome**, prompting urgent patch development.
- Executed regression testing across 4 major WebOS versions (v5, v6, v22, v23), achieving a **99.5%** pass rate for critical path stability.

Data Science Intern *Zidio Development* Jan 2025 – Apr 2025

- Architected an **Employee Emotion Analysis System** utilizing **Computer Vision** (Facial Recognition) and **NLP**, achieving **94%** classification accuracy.
- Implemented privacy-focused data pipelines to process real-time video/audio streams while ensuring compliance with employee data confidentiality standards.
- Integrated the model with a scalable backend to provide real-time stress alerts and task recommendations.

Projects

MitraBot (Generative AI) [GitHub](#) 

- Fine-tuned a **Llama 3.1** model to handle 50+ distinct government service queries, utilizing advanced **Named Entity Recognition (NER)**.
- Benchmark Achievement:** Outperformed **GPT-4** and **Gemini 1.5** in specific context-retention and reasoning tasks regarding local government schemes.
- Engineered multi-modal support for text, voice, and regional languages using **Hugging Face** transformers.

EmotiSense AI [GitHub](#) 

- Developed a real-time multimodal emotion detection system using **TensorFlow CNNs** (Visual) and **Librosa** (Audio).
- Achieved **95%** accuracy in identifying user emotional states to drive an intelligent task recommendation engine.
- Deployed via **FastAPI** with a **Streamlit** dashboard for real-time inference visualization.

Smart Router & Network Guardian [GitHub](#) 

- Architected a security-focused network solution on Raspberry Pi featuring **VPN** and **Firewall** capabilities.
- Implemented **DNS Sinkholing** using Pi-hole, blocking over **30%** of malicious web traffic network-wide.
- Integrated **Grafana** and **Prometheus** for real-time traffic monitoring and analytics.

Apartment Management System [GitHub](#) 

- Constructed a full-stack application using **Angular** and **Node.js** that reduced administrative data entry time by an estimated **40%**.

Education

Presidency University *B.Tech in Computer Science and Engineering* Aug 2021 – July 2025

- GPA:** 7.5/10.0
- Relevant Coursework:** Data Structures, Algorithms, Operating Systems, DBMS, Computer Networks, Machine Learning, Deep Learning, OOP, Applied Statistics, Software Engineering.

Technical Skills

- AI & Data Science:** Generative AI, Llama 3.1, Machine Learning, NLP, Computer Vision, TensorFlow, OpenCV, Pandas, NumPy, SQL
- Languages & Backend:** Python, Java, C, FastAPI, Data Structures, OOP, System Performance
- Infrastructure & DevOps:** Linux, Git, Docker, Grafana, Prometheus, Raspberry Pi, WebOS, Bash
- Web & QA:** Angular, JavaScript, HTML5, Quality Assurance, Regression Testing

Publications

- EmotiSense AI: Multimodal Emotion Detection Platform** [🔗](#) Mar 2025
Adithya V, Pakruddin B
- Detailed the architecture for real-time holistic emotion analysis using Vision and Speech Processing.
- MitraBot: Guide towards Loans and Schemes** [🔗](#) Jan 2025
Sathvik NB Math, Adithya V, et al.
- Comparative analysis demonstrating superior context retention over generalist LLMs for niche government data.

Honors & Certifications

- State Level Taekwondo Championship** Silver Medalist
- Competed in the state-level championship (Mysore); demonstrated discipline and strategic thinking under pressure.
- Data Structures and Algorithms using C** Jun 2022
- Certification demonstrating proficiency in implementing core data structures and performance analysis.